

CS & IT

Theory of computation

DPP: 1

Context Free Languages

Q1 Consider the following grammar:

$$S \rightarrow abSba \mid bSb \mid aSa \mid a \mid b \mid \epsilon$$

What is the language generated by above grammar?

- (A) CFG that generates all palindromes over alphabet (a, b).
- (B) CFG that generates all palindromes over alphabet (a, b) that do not contain substring "ab".
- (C) CFG that generates all palindromes over alphabet (a, b) that contain substring "aa".
- (D) CFG that generates all palindromes over alphabet (a, b) that do not contain substring "aa".

Q2 Consider the following grammars:

$$G_1: S \rightarrow aSc \mid \$ B$$

$$B \rightarrow bBc \mid \$$$

$$G_2: S \rightarrow aaSc \mid \$ B$$

$$B \rightarrow bBc \mid \$$$

$$G_3: S \rightarrow aSc \mid \$ B$$

$$B \rightarrow bbBcc \mid \$$$

Which of the following is true regarding G_1 , G_2 and G_3 ?

- (A) $L(G_1) \subseteq L(G_2), L(G_2) \subseteq L(G_3)$
- (B) $L(G_2) \subseteq L(G_3), L(G_1) \subseteq L(G_3)$
- (C) $L(G_1) \subseteq L(G_3), L(G_3) \subseteq L(G_2)$
- (D) $L(G_2) \subseteq L(G_1), L(G_3) \subseteq L(G_1)$

Q3 Consider the following context free grammar:

$$S \rightarrow aA \mid aBB$$

$$A \rightarrow aaA \mid \epsilon$$

$$B \rightarrow bB \mid bbC$$

$$C \rightarrow B$$

What will be the equivalent simplified CFG for the given grammar?

$$(A) S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa \mid b$$

$$(B) S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa$$

$$(C) S \rightarrow aAa \mid B$$

$$A \rightarrow aaA \mid aa$$

$$B \rightarrow bB \mid bb$$

$$(D) S \rightarrow aAa \mid B$$

$$A \rightarrow aA \mid b$$

$$B \rightarrow bB \mid bb \mid a$$

Q4 Consider the following grammar:

$$S \rightarrow AB$$

$$A \rightarrow BaB \mid a$$

$$B \rightarrow bbA$$

Which of the following is true regarding given grammar?

- (A) Every string of the above grammar have at least two a's.
- (B) Every string have three consecutive a's.
- (C) Every string have alternate a and b.
- (D) Every string have b's in multiple of 2.

Q5 Given the following two grammars:

$$G_1: S \rightarrow AB \mid aaB$$

$$A \rightarrow a \mid Aa$$

$$B \rightarrow b$$

$$G_2: S \rightarrow aSbS \mid bSaS \mid \epsilon$$

What is true regarding above grammars?

- (A) Both G_1 and G_2 are ambiguous.



- (B) Both G_1 and G_2 are unambiguous.
 (C) Only G_1 is ambiguous.
 (D) Only G_1 is unambiguous.

Q6 Consider the following languages

$$L_1 = \{a^n b^{n+m} c^m \mid n, m \geq 0\}$$

$$L_2 = \{a^p b^q c^{p+q} \mid p, q \geq 0\}$$

Which of the following is true?

- (A) Only L_1 is Regular.
 (B) Only L_2 is Regular.
 (C) Both L_1 and L_2 are Regular.
 (D) None of these

Q7 Which of the following is string accepting mechanism of PDA?

- (A) PDA using final state.
 (B) PDA using empty stack.
 (C) PDA using both empty stack and final state.
 (D) PDA using transition state.

Q8 Which of the following is correct push operation:

- (A) $\Sigma(q, a, b) = (q', ab)$
 (B) $\delta(q, a, b) = (q, ab)$
 (C) $\delta(q, a, b) = (q', ab)$
 (D) $\Sigma(q, a, b) = (q', \epsilon)$

Q9 Consider the following statements:

- (I) All DPDA are NPDA.
 (II) All NPDA are DPDA.
 (III) All NPDA and DPDA are equivalent.
 (IV) All context free language are regular language.

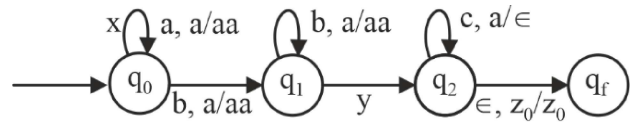
The total number of correct statements are_____.

Q10 What does following transition means:

$$\delta(q, \epsilon, b) = (q', b)$$

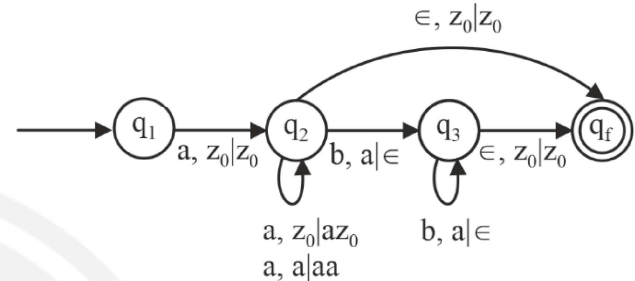
- (A) Push b
 (B) Pop b
 (C) Read b
 (D) No operation

Q11 What are the values of x and y, if the language accepted by NPDA is $L = \{a^m b^n c^{(m+n)} \mid m, n \geq 1\}$.



- (A) $x = \epsilon, a/a$; $y = c, b/\epsilon$
 (B) $x = a, z_0/z_0$; $y = c, b/c$
 (C) $x = a, a/aa$; $y = b, c/c$
 (D) $x = a, z_0/az_0$; $y = c, a/\epsilon$

Q12 Which language is accepted by the following PDA.

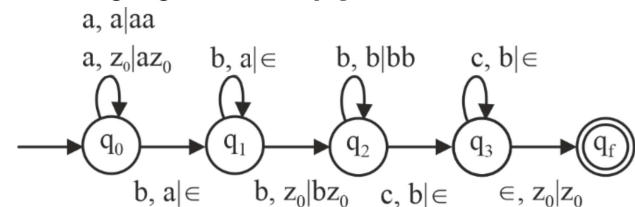


- (A) $L = \{a^n b^{n+1} \mid n \geq 1\}$
 (B) $L = \{a^{n+1} b^m \mid n, m \geq 1\}$
 (C) $L = \{a^{n+1} b^n \mid n \geq 0\}$
 (D) $L = \{a^n b^{n+1} \mid n \geq 0\}$

Q13 Which of the following languages are accepted by PDA.

- (A) $L = \{a^n b^n c^m \mid m, n \geq 1\}$
 (B) $L = \{a^n b^n c^m \mid m \leq n\}$
 (C) $L = \{a^n b^m c^n d^m \mid m, n \geq 0\}$
 (D) $L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$

Q14 The language derived by given PDA is



- (A) $L = \{a^n b^m c^k \mid n = m + k, m, n, k \geq 0\}$.
 (B) $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 1\}$.
 (C) $L = \{a^m b^n c^k \mid k = m + n, m, n, k \geq 0\}$.
 (D) $L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}$.

Q15



Suppose L_1 is a finite language and L_2 is non-regular language then $L_1 \cap L_2$ will be:

- (A) Regular but infinite.
- (B) Non-regular.
- (C) Finite and regular.
- (D) None of these.

Q16 Consider following statements:

S_1 : Kleene Closure of infinite set is always finite.

S_2 : Kleene Closure of finite set is always infinite.

Which of the following is correct?

- (A) S_1 only
- (B) S_2 only
- (C) Both S_1 and S_2
- (D) None of these

Q17 Consider a language $L (a+b)^*$, then subset of L will be?

- (A) Regular
- (B) Regular but finite
- (C) Non-regular
- (D) None of these

Q18 Let L_1 be a regular language and L_2 be a CFL language, then which of the following is correct?

- (A) $L_1 \cap L_2$ is CFL
- (B) $L_1 - L_2$ must be regular
- (C) $L_1 \subset L_2$ must be regular
- (D) None of these.

Q19 Consider two languages L_1 and L_2 .

$L_1 \subset L_2$, L_2 is regular then which of the following is true?

- (A) L_1 must be finite
- (B) L_1 must be regular
- (C) L_1 must be non-regular
- (D) None of these

Q20 Regular language does not closed under on which operation?

- (A) Complement
- (B) Union
- (C) Subset
- (D) Intersection.



Answer Key

Q1 (A)

Q2 (D)

Q3 (B)

Q4 (A, D)

Q5 (A)

Q6 (D)

Q7 (A, B, C)

Q8 (B, C)

Q9 1

Q10 (D)

Q11 (D)

Q12 (C)

Q13 (A, D)

Q14 (D)

Q15 (C)

Q16 (D)

Q17 (D)

Q18 (A)

Q19 (D)

Q20 (C)



Hints & Solutions

Q1 Text Solution:

Given

$$S \rightarrow abSba \mid bSb \mid aba \mid a \mid b \mid \hat{1}$$

Strings formed using above grammar are:

$\{\epsilon, a, b, aa, bb, \dots\}$

This is the CFG that generates all palindromes over alphabet (a, b) .

So, option (a) is correct answer.

Q2 Text Solution:

$$L(G_1) = \{a^m b^n c^{m+n} \mid m, n \geq 0\}$$

$$L(G_2) = \{a^{2m} b^n c^{2m+n} \mid m, n \geq 0\}$$

$$L(G_3) = \{a^m b^{2n} c^{m+2n} \mid m, n \geq 0\}$$

$\therefore$ It is clear,

$$L(G_2) \subseteq L(G_1), \text{ and}$$

$$L(G_3) \subseteq L(G_1)$$

Therefore, option d is correct.

Q3 Text Solution:

Simplified CFG is a CFG without any null productions, unit-productions and useless symbol.

1. Elimination of NULL Productions:

$$A \rightarrow \epsilon$$

Keep all the productions as it is and substitute

ϵ in place of A.

$$S \rightarrow aA \mid aBB \mid a$$

$$A \rightarrow aaA \mid aa$$

$$B \rightarrow bB \mid bbC$$

$$C \rightarrow B$$

2. Eliminate Unit Productions: If the productions is of the form $A \rightarrow B$, the production is known as unit production. Eliminate unit production by replacing equivalent derivations.

$$S \rightarrow aA \mid aBB \mid a$$

$$A \rightarrow aaA \mid aa$$

$$B \rightarrow bB \mid bbB$$

3. Eliminate useless symbol: The variables that are not involved the derivation of any string is known as useless symbol.

In this question B is deriving $bB \mid bbB$ which is entering into infinite loop. So, eliminating all productions of B.

$$S \rightarrow aA \mid a$$

$$A \rightarrow aaA \mid aa$$

Hence, options (b) is the correct answer.

Q4 Text Solution:

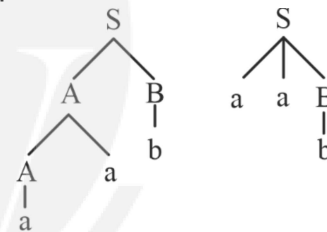
The string produced by given grammar is $\{abba, abbaabba, \dots\}$

So option (a) and (d) are the true statements.

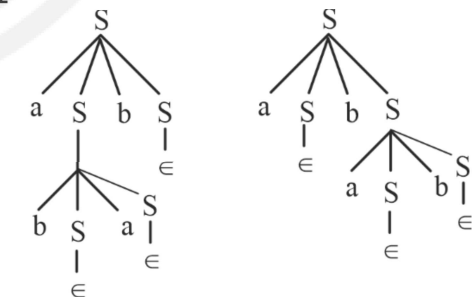
Q5 Text Solution:

Both G_1 and G_2 are ambiguous, as both have multiple parse tree for same string.

G_1 : aab



G_2 : abab



Q6 Text Solution:

$$L_1 = \{\epsilon, ab, bc, aabb, bbcc, \dots\}$$

$$L_2 = \{\epsilon, ac, bc, aacc, bbcc, \dots\}$$

Both the languages are CFL not regular.

Therefore option (d) is correct.



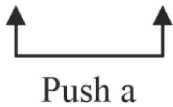
Q7 Text Solution:

PDA accepts string using final state, empty stack & both and all methods are equivalent.

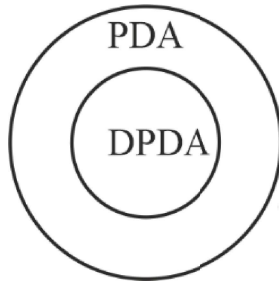
Q8 Text Solution:

$\delta(q, a, b) = (q, ab)$ is correct push operation on current state.

$\delta(q, a, b) = (q', ab)$ is the correct push operation
 $\delta(q, a, b) = (q', ab)$

**Q9 Text Solution:**

(I) All DPDA are PDA: correct



(II) All PDA are DPDA: incorrect

Some PDA are DPDA but all PDA are not DPDA.

(III) All PDA and DPDA are equivalent: incorrect

DPDA are subset of PDA. So all PDA and DPDA are not equivalent.

(IV) All context free language are regular language: incorrect

CFL are superset of RL. So, this statements is incorrect. However, all regular languages are context free languages.

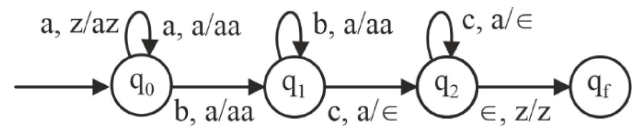
Therefore only 1 statement is correct.

Q10 Text Solution:

$\delta(q, \epsilon, b) = (q', b)$, this represents no operation is performed on stack. So, there is no change in the stack.

Q11 Text Solution:

The PDA will be:



The x is to insert/push initial “a” into stack.

The y is to remove/pop an “a” for each “c”.

Q12 Text Solution:

$$L = \{a^{n+1} b^n \mid n \geq 0\}$$

$$= \{a^1 a^n b^n \mid n \geq 0\}$$

↓ ↓ ↓

Skip Push Pop

In the given PDA, first “a” is being skip. Then, for “n” number of ‘a’ are inserted and once ‘b’ starts appearing for each ‘b’, we will pop one “a”.

Hence, option (c) is correct.

Q13 Text Solution:

(a) $L = \{a^n b^n c^m \mid m, n \geq 1\}$

Push ‘n’ no of ‘a’. pop ‘a’ for each ‘b’, skip ‘c’.

It is accepted by PDA.

(b) $L = \{a^n b^n c^m \mid m \leq n\}$

Push ‘n’ no of ‘a’, pop ‘a’ for each ‘b’. Since, it is given that $m \leq n$, so we can’t compare number of c. Therefore, it is not acceptable using PDA.

(c) $L = \{a^n b^m c^k d^m \mid m, n \geq 0\}$

Unable to compare.

Therefore, it is not accepted by PDA.

(d) $L = \{a^m b^n c^n d^m \mid m, n \geq 0\}$

Push ‘m’ no. of “a”. Push ‘n’ no. of “b”.

Pop ‘b’ for each ‘c’. Pop ‘a’ for each ‘d’.

Hence, it is accepted by PDA.

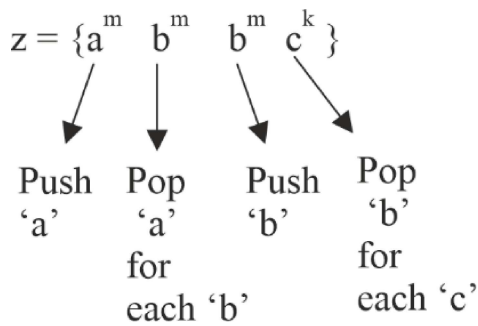
Q14 Text Solution:

The given PDA will generate language

$$L = \{a^m b^n c^k \mid n = m + k, m, n, k \geq 1\}$$

$$L = \{a^m b^{m+k} c^k\}$$





Hence, option (d) is correct.

Q20 Text Solution:

Subset of regular language need not be regular

Q15 Text Solution:

Finite $\cap$ non-regular always finite.

Hence, option (c) is correct.

Q16 Text Solution:

S₁: False

Set = $\{\epsilon\} = \{\epsilon\}^* = \epsilon$ only (Finite)

S₂: Set = $\{a\} = \{a\}^* = \epsilon, a, aa, aaa, \dots = (a^*)$ (Infinite)

So, both statements are false.

Hence, option (d) is correct.

Q17 Text Solution:

$(L) = (a + b)^*$

- $a^n b^n$ is a subset of $(a + b)^*$
but $a^n b^n$ is not a regular and also not finite.
- ab is a subset of L but ab is a finite and regular.

Hence, option (d) is correct.

Q18 Text Solution:

$L_1 = \text{Regular}$

$L_2 = \text{CFL}$

(a) Regular $\cap$ CFL Need not be regular.

Example:

$= (a + b)^* (L_1) \cap a^n b^n (L_2)$

$= a^n b^n (\text{CFL}).$

(b) Regular – CFL

$= \text{Regular} \cap \overline{\text{CFL}}$

$= \text{Need not be regular}$

(c) Regular $\subset$ CFL

Need not be regular

Hence, option (d) is correct.

Q19 Text Solution:

If $L_1 \subset L_2$, then L_1 can be finite or infinite

